

Key words and Terms

thermodynamics	热力学
quasistatic process	准静态过程
molar specific heat	摩尔热容
isothermal process	等温过程
isobaric process	等压过程
isochoric process	等容过程
adiabatic process	绝热过程
monatomic molecules	单原子分子
diatomic molecules	双原子分子
polyatomic molecules	多原子分子
degrees of freedom	自由度
principle of equipartition of energy	能量均分原理
internal energy	内能

Chapter 14 The first Law of Thermodynamics

14.1 Heat, Work and Internal Energy

14.2 The first Law of Thermodynamics

14.3 Applying the First Law of Thermodynamics

Chapter 14 The first Law of Thermodynamics

System(系统): Objects that we want to consider

Environment(外界): everything else in the universe
系统以外的物体

Categories of system

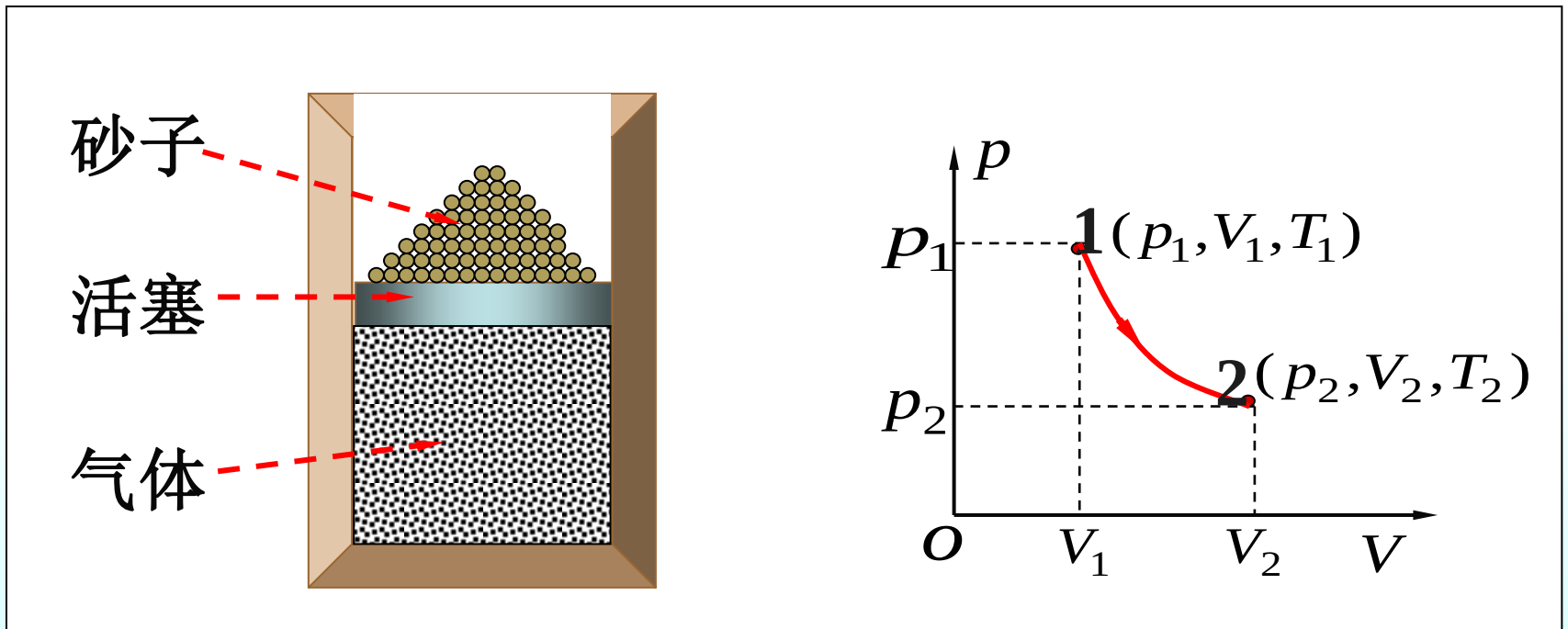
Isolated system(孤立系统): 与外界没有任何相互作用的系统既没能量交换，也没物质交换。是一个理想模型。

Closed system(封闭系统): 只有能量交换, 没有物质交换。

Open system(开放系统): 不但有能量交换, 而且还进行物质交换。

Quasistatic (准静态) process: a process in which every step goes slowly that it is never far away from **an equilibrium state**.

And it can be represented by a smooth curve on the **pressure-volume diagram**. (pV 图)



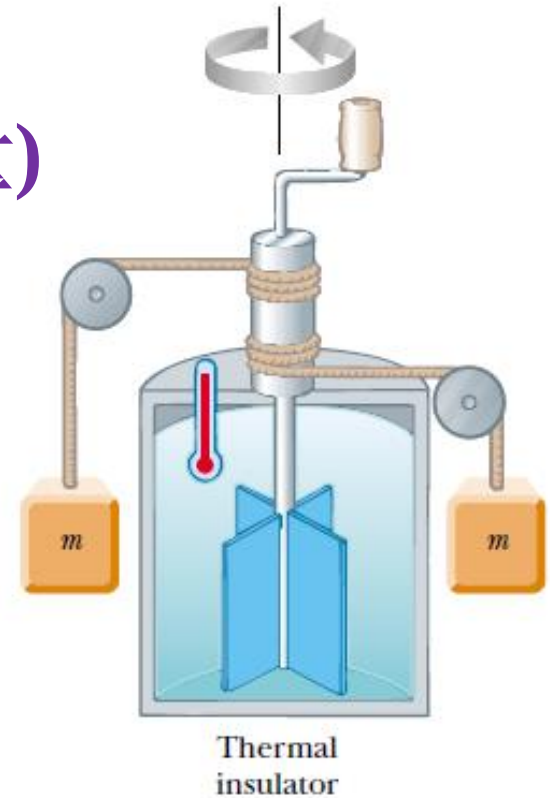
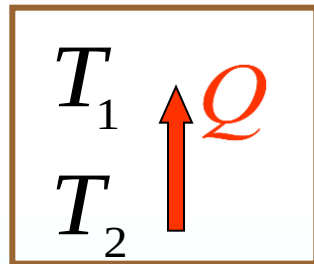
14.1 Heat, Work and Internal Energy

1. Heat (process variable 过程量)

$$1 \text{ Calorie} = 4.186 \text{ J}$$

mechanical equivalent of heat

$$T_1 < T_2$$

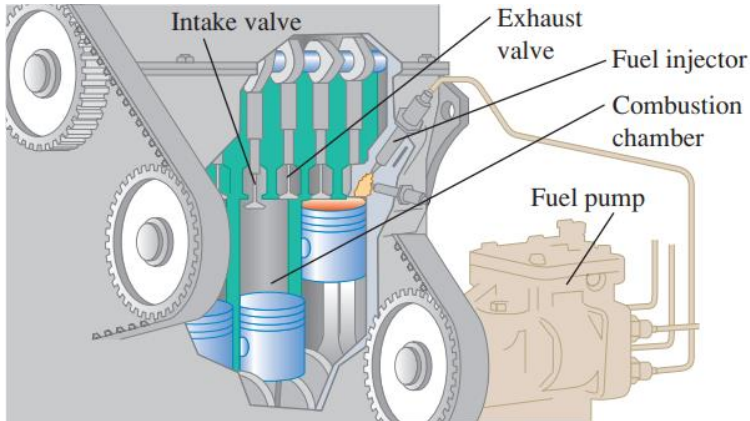


Heat is energy that is transferred from one body to another because of a difference in **temperature**.

通过传热方式传递能量的量度，系统和外界之间存在**温差**而发生的能量传递。

2. Work done in volume changes (process variable)

Cutaway of an automobile engine.



$$dW = Fdx = pAdx$$

$$= pAdx = pdV$$

$$W = \int_{V_1}^{V_2} p dV$$

(work done in a volume change)

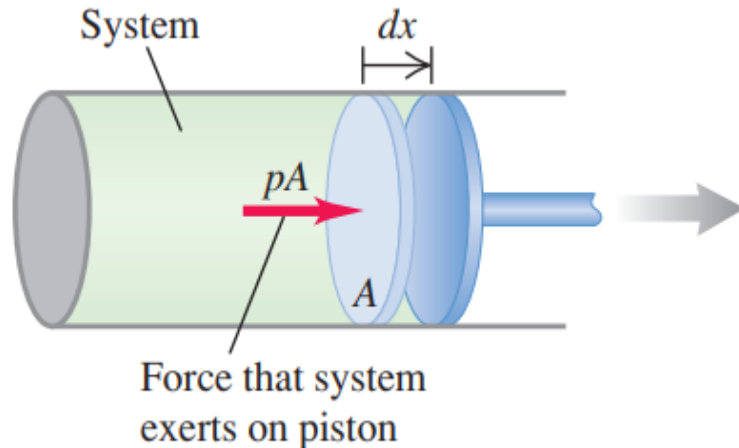
$W > 0$, expansion, work done by

the gas on the environment

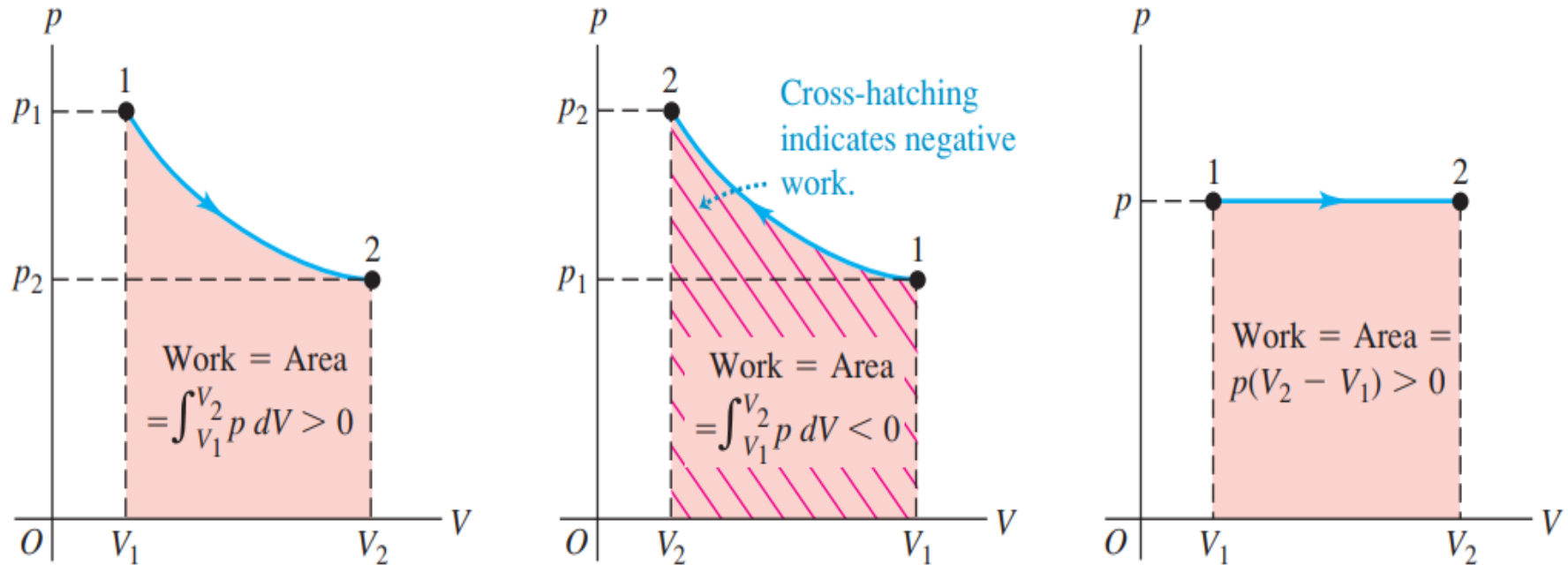
系统对外界作正功

$W < 0$, compression, work done by
the environment on the gas

外界对系统作正功



2. Work done in volume changes (process variable)



Work equals area under the curve in a PV diagram.

The work done in taking a system from one state to another depends not only on the initial and final states but also on the **path** or process.

3. The internal energy of ideal gas U

Internal energy U : the sum of all the energy of all the molecules in an object.

- Energy within molecules

- Kinetic energy of atoms**
 - Potential energy between atoms = 0 (ideal gas)**

- Potential energy between molecules = 0 (ideal gas)

Internal energy is all of a system's **kinetic energy** of translation, rotation and vibration of molecules.

(1) Degrees of Freedom (DF)

DF 1: The **number** of independent velocity components (either linear or angular) needed to describe the **motion** of a molecule completely

DF 2: The **number** of independent ways molecules can possess (store) **energy**

$$f = t + r + v$$

t: translational motion

r: rotational motion

v: vibration

(2) DF of ideal gas for the three kinds of molecules

(a) **monatomic** molecules-contain only a single atom, as He (Helium)

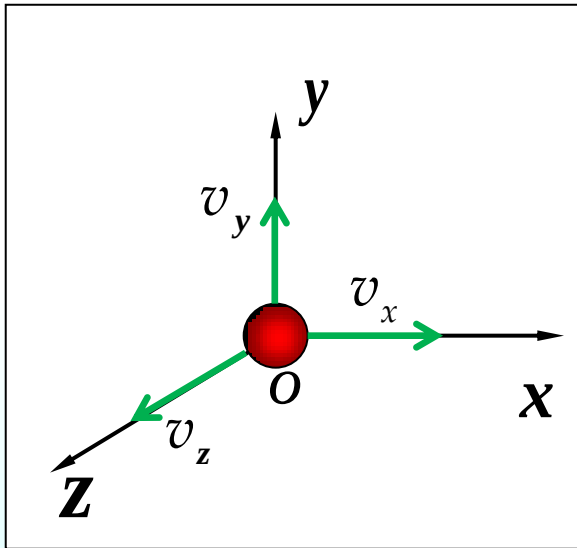
Three degrees of **translational** freedom

$$v_x, v_y, v_z$$

$$f = t = 3$$

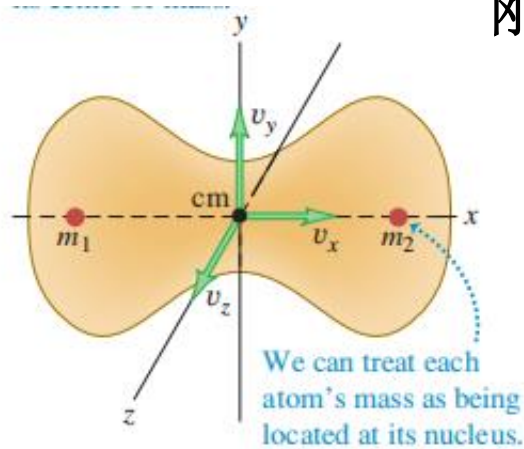
$$\bar{K} = \frac{1}{2} m \overline{v_x^2} + \frac{1}{2} m \overline{v_y^2} + \frac{1}{2} m \overline{v_z^2} = \frac{3}{2} kT$$

$$= 3 \left(\frac{1}{2} kT \right)$$



(b) Diatomic molecules-contain two atoms, as O_2

The **center** of mass of the molecule can translate in the x , y , and z directions. In addition, the molecule can **rotate** about three mutually perpendicular axes.



刚性分子：不考虑分子间原子的振动，只考虑平动加转动

Three degrees of **translational** freedom

v_x , v_y , v_z of cm

Two degrees of **rotational** freedom

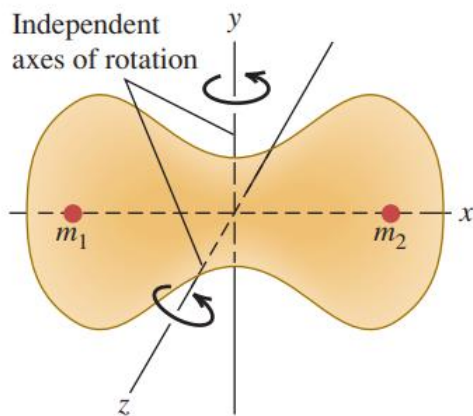
ω_y , ω_z

$$f = t + r = 3 + 2 = 5$$

$$\bar{K}_t = \frac{1}{2} m \overline{v_{Cx}^2} + \frac{1}{2} m \overline{v_{Cy}^2} + \frac{1}{2} m \overline{v_{Cz}^2} = 3 \left(\frac{1}{2} kT \right)$$

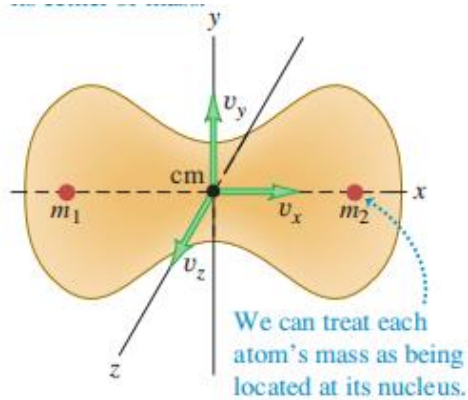
$$\bar{K}_r = \frac{1}{2} I \overline{\omega_y^2} + \frac{1}{2} I \overline{\omega_z^2} = 2 \left(\frac{1}{2} kT \right)$$

$$\bar{K}_{\text{total}} = \bar{K}_t + \bar{K}_r = 5 \left(\frac{1}{2} kT \right)$$



(b) Diatomic molecules-contain two atoms, as O_2

* 非刚性双原子气体分子：考虑分子间原子的振动



Three degrees of **translational** freedom

v_x , v_y , v_z of cm

Two degrees of **rotational** freedom

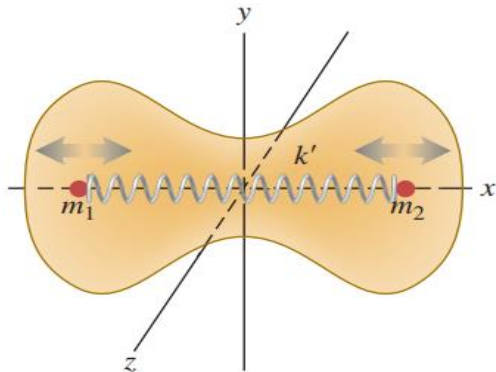
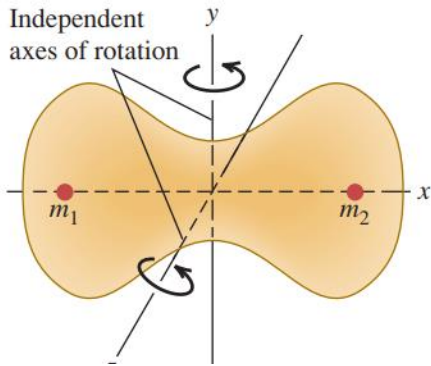
ω_y , ω_z

One degree of **vibrational** freedom

$$f = t + r + v = 3 + 2 + 1 = 6$$

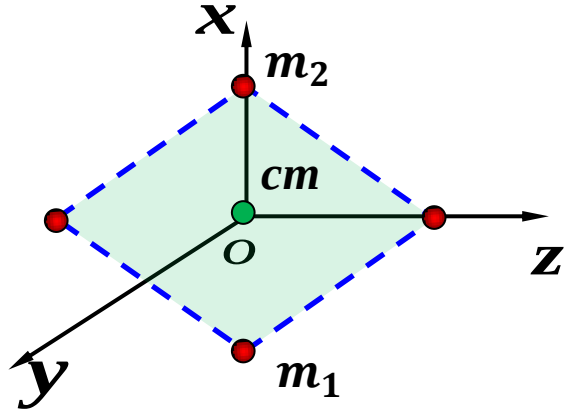
$$\bar{K}_{\text{total}} = \bar{K}_t + \bar{K}_r + \bar{K}_v$$

$$= 3\left(\frac{1}{2}kT\right) + 2\left(\frac{1}{2}kT\right) + 2\left(\frac{1}{2}kT\right) = 7\left(\frac{1}{2}kT\right)$$



(c) Polyatomic molecules-contain three or more atoms

刚性分子：不考虑分子间原子的振动，只考虑平动加转动



Three degrees of **translational** freedom

v_x , v_y , v_z of cm

Three degrees of **rotational** freedom

ω_y , ω_z , ω_x

$$f=t+r=3+2+1=6$$

$$\bar{K}_t = \frac{1}{2} m \overline{v_{Cx}^2} + \frac{1}{2} m \overline{v_{Cy}^2} + \frac{1}{2} m \overline{v_{Cz}^2} = 3 \left(\frac{1}{2} kT \right)$$

$$\bar{K}_r = \frac{1}{2} I \overline{\omega_y^2} + \frac{1}{2} I \overline{\omega_z^2} + \frac{1}{2} I \overline{\omega_x^2} = 3 \left(\frac{1}{2} kT \right)$$

$$\bar{K}_{\text{total}} = \bar{K}_t + \bar{K}_r = 6 \left(\frac{1}{2} kT \right)$$

(3) The principle of equipartition of energy

The principle of equipartition of energy : energy is shared **equally** among the **active degrees** of freedom, and in particular each active degree of freedom of a molecule has on the average an **energy** equal to $\frac{1}{2}kT$.

分子任何一个**自由度**的平均能量都相等，均为 $\frac{1}{2}kT$

Monatomic gas $U = 3\left(\frac{1}{2}kT\right) \times N = \frac{3}{2}NkT = \frac{3}{2}nRT$

Diatomic gas $U = 5\left(\frac{1}{2}kT\right) \times N = \frac{5}{2}NkT = \frac{5}{2}nRT$

Polyatomic gas $U = 6\left(\frac{1}{2}kT\right) \times N = \frac{6}{2}NkT = 3nRT$

Internal energy

$$U = \frac{f}{2} nRT = \frac{f}{2} NkT$$

internal energy of **one mol of an idea gas**

$$U = \frac{f}{2} RT$$

internal energy of **n mol of an idea gas**

$$U = \frac{f}{2} nRT$$

the change in internal energy

$$\Delta U = \frac{f}{2} nR\Delta T$$

The internal energy is a **state function of **T**. It depends only on the initial and final states and does not depend at all on how the system gets from one to the other.**

14.2 The first Law of Thermodynamics

The change in internal energy of a **closed system** will be equal to the **heat added to the system** **minus** the **work done by the system**.

$$\Delta U = Q - W$$

$W > 0$, Work by system

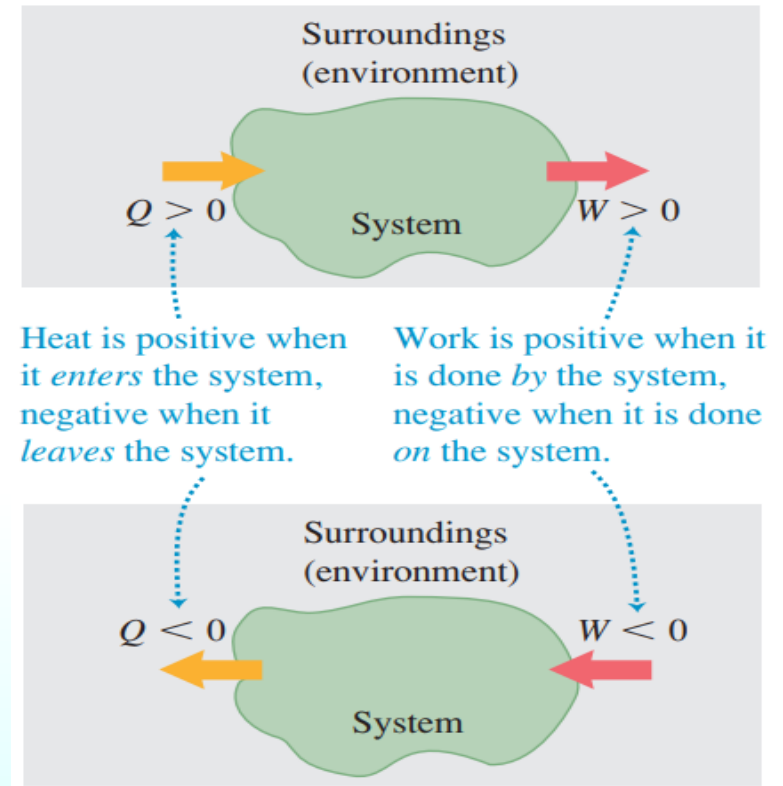
$W < 0$, Work on system

$Q > 0$, Heat added

$Q < 0$, Heat lost

$\Delta U = U_2 - U_1$ the change in internal energy(内能的增量)

$T \uparrow, \Delta U > 0, \quad T \downarrow, \Delta U < 0,$



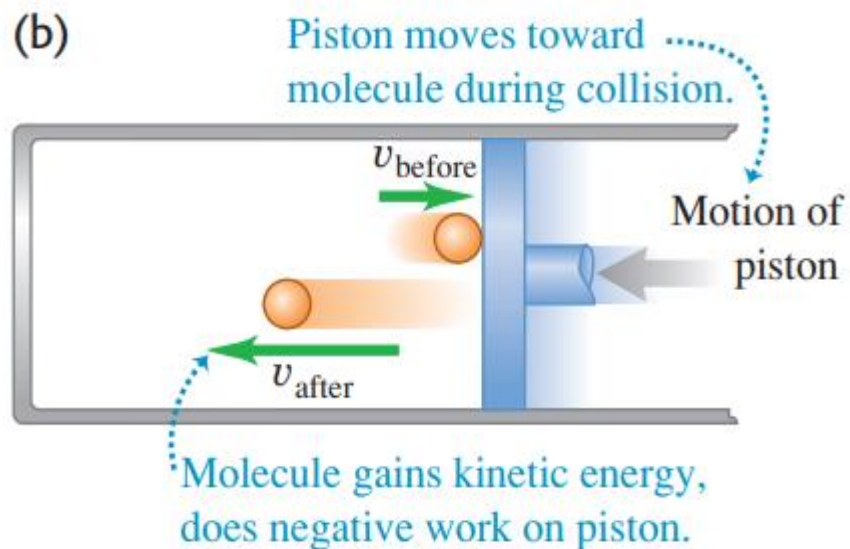
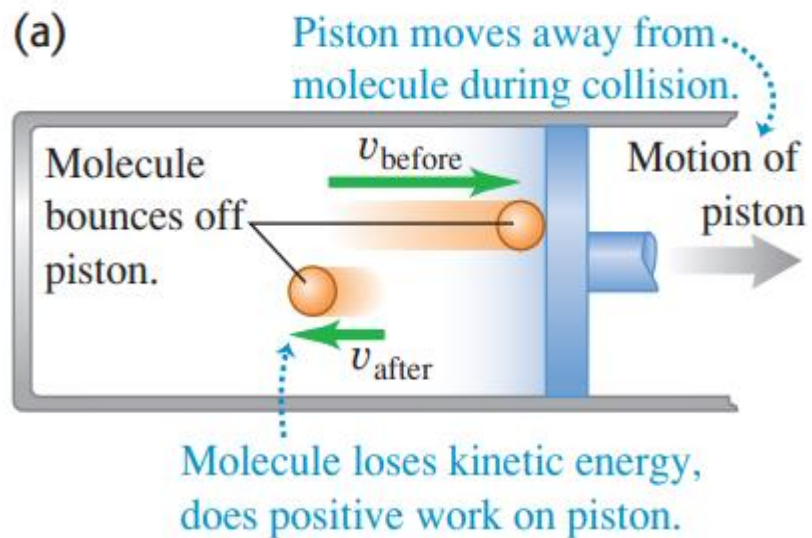
$$dU = dQ - dW \quad \text{Differential form}$$

(1)热力学第一定律就是**能量转化与守恒定律**，它是自然界中的一个普遍规律。

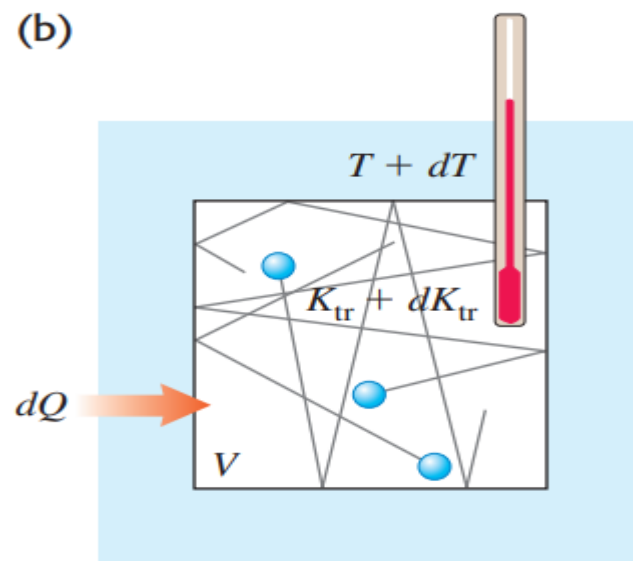
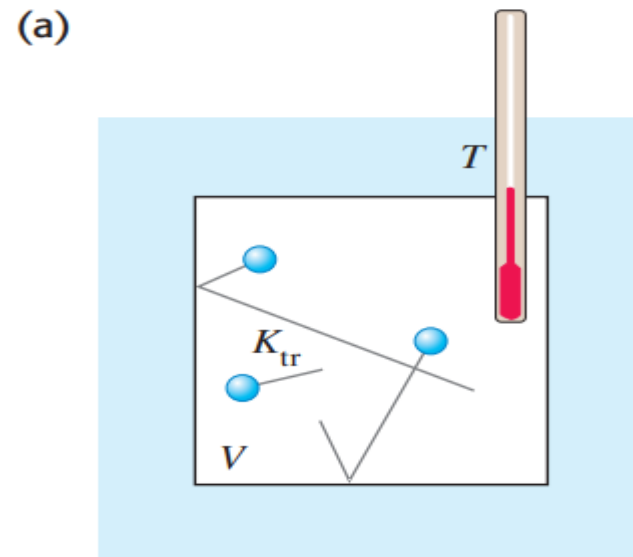
(2)系统状态变化过程中，功与热之间的转换**不可能**是直接的，总是通过物质系统来完成。向系统传递热量，使系统内能增加，再由系统内能减少来对外做功；或者外界对系统做功，使系统内能增加，再由内能减少，系统向外界传递能量：



Work $\rightarrow$ change in internal energy



Heat $\rightarrow$ change in internal energy



Applying the first law of thermodynamics

Q	W	ΔU	Possible?
+	+	+	
—	—	—	
+	+	—	
+	—	+	
+	—	—	
—	—	+	
—	+	—	
—	+	+	

例：一理想气体系统由状态a沿acb到达状态b，有350J热量传入系统，而系统做功 130J.

(1) 经过adb过程，系统做功40J，传入系统的热量 $Q = ?$

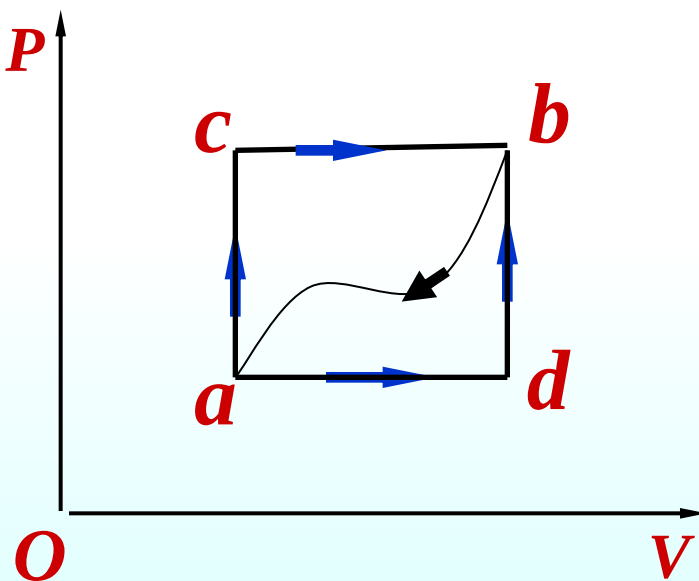
(2) 当系统由状态b沿曲线ba返回状态a时，外界对系统功 60J，则系统吸收热量 $Q = ?$

$$\Delta U_{ab} = Q_{acb} - W_{acb} = 350 - 130 = 220J$$

$$Q_{acb} = \Delta U_{ab} + W_{acb} = 220 + 40 = 260J$$

$$\Delta U_{ba} = -220J$$

$$Q_{ba} = \Delta U_{ba} + W_{ba} = -220 + (-60) = -280J$$

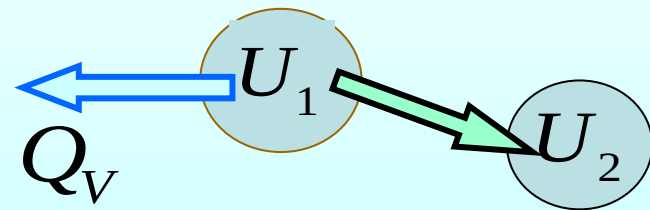
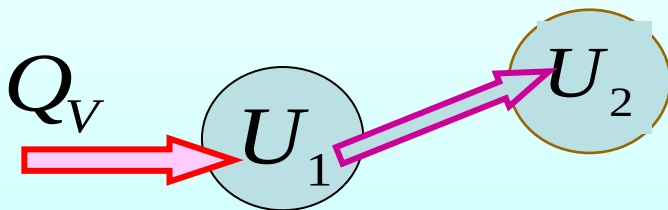
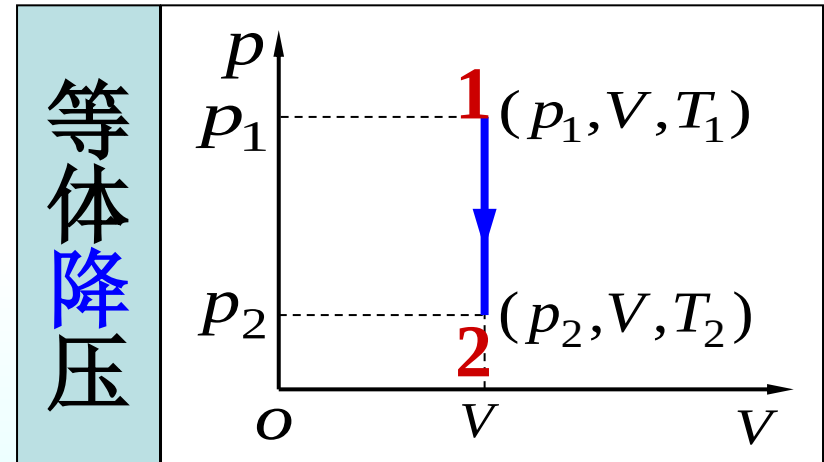
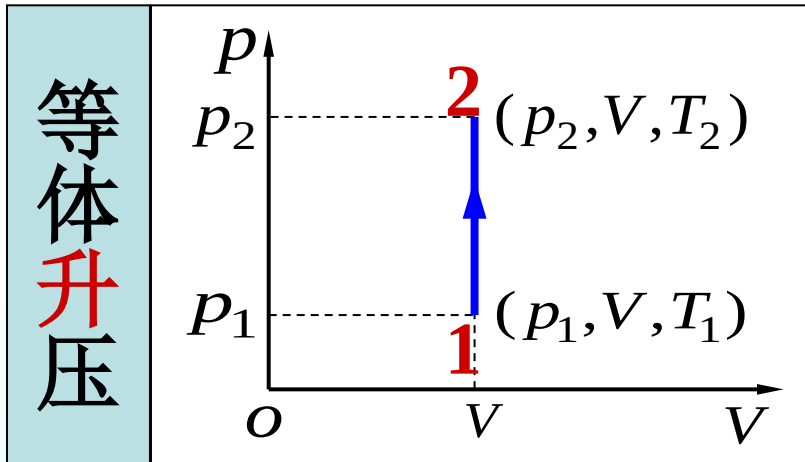


14.3 Applying the First Law of Thermodynamics

1. Isochoric Processes(等体过程)

(1) Process equation (Restriction)过程方程

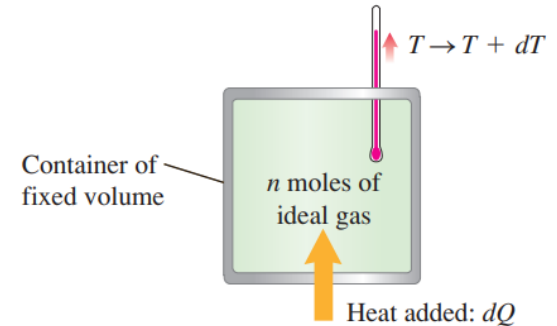
(2) pV diagram $\frac{p}{T} = \text{const}$



(a) Constant volume: $dQ = nC_V dT$

$$(3) \quad W = 0$$

$$Q_V = \Delta U = n \frac{f}{2} R \Delta T$$



(4) C_V **molar specific heat at constant volume**
摩尔定体热容 unit: $\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

The heat required to raise 1 mol of the gas by 1°C at constant volume.

$$C_V = \frac{dQ_V}{dT} \quad dQ_V = C_V dT$$

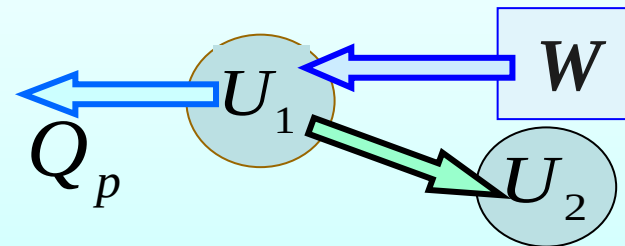
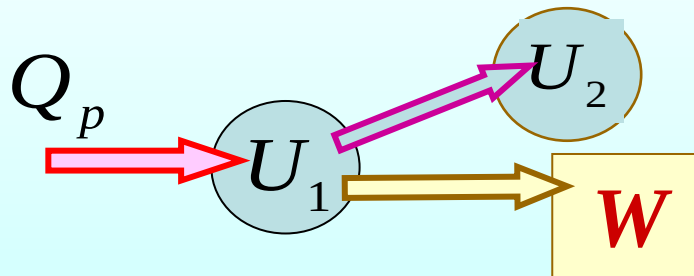
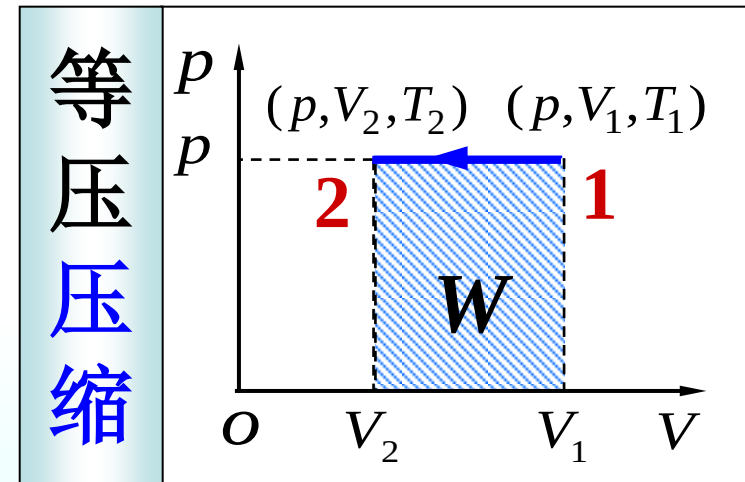
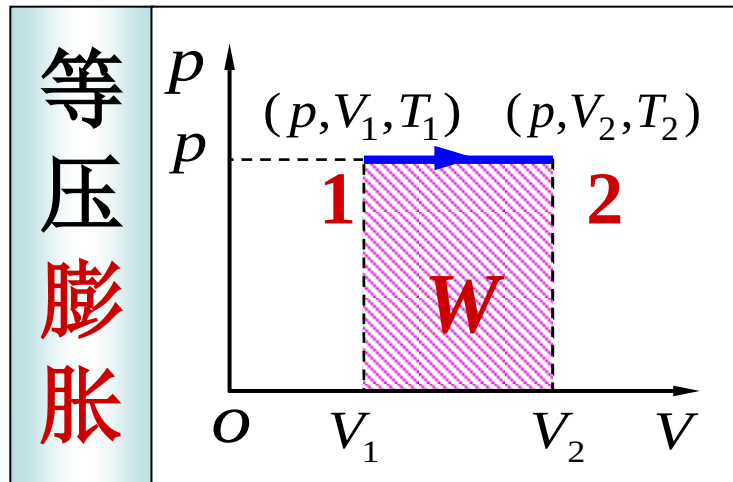
$$nC_V \Delta T = n \frac{f}{2} R \Delta T$$

$$C_V = \frac{f}{2} R$$

2. Isobaric Processes (等压过程)

(1) Process equation $\frac{V}{T} = \text{constant}$

(2) pV diagram



$$(3) \quad W = p(V_2 - V_1) = nRT_2 - nRT_1 = nR\Delta T$$

$$\Delta U = n \frac{f}{2} R\Delta T$$

$$Q_p = \Delta U + W = n \frac{f}{2} R\Delta T + nR\Delta T = n\left(\frac{f}{2} + 1\right)R\Delta T$$

(4) C_p **molar specific heat at constant pressure**
 摩尔定压热容 unit: $\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

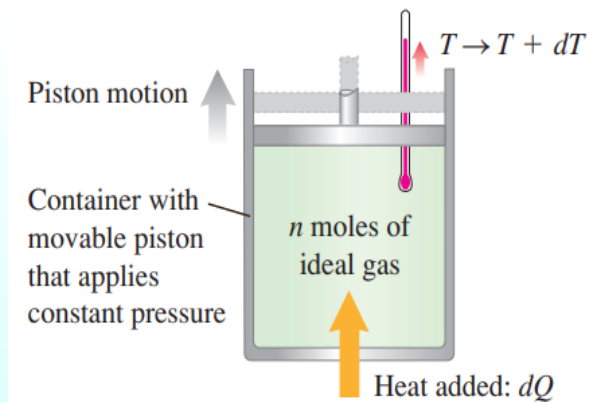
The heat required to raise 1 mol of the gas by 1°C at constant pressure

$$C_p = \frac{dQ_p}{dT} \quad dQ_p = C_p dT$$

$$nC_p \Delta T = n\left(\frac{f}{2} + 1\right)R\Delta T \quad C_p = \left(\frac{f}{2} + 1\right) R$$

$$(5) \gamma = \frac{C_p}{C_v} = \frac{f+2}{f} \text{ (比热容比)} \quad C_p = C_v + R \text{ (迈耶公式)}$$

(b) Constant pressure: $dQ = nC_p dT$

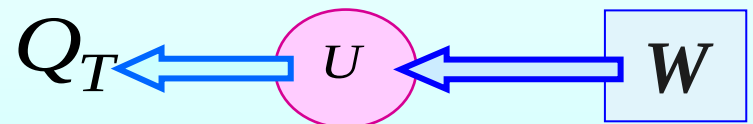
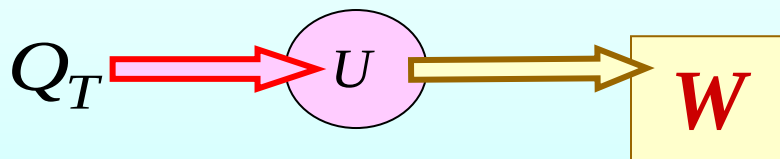
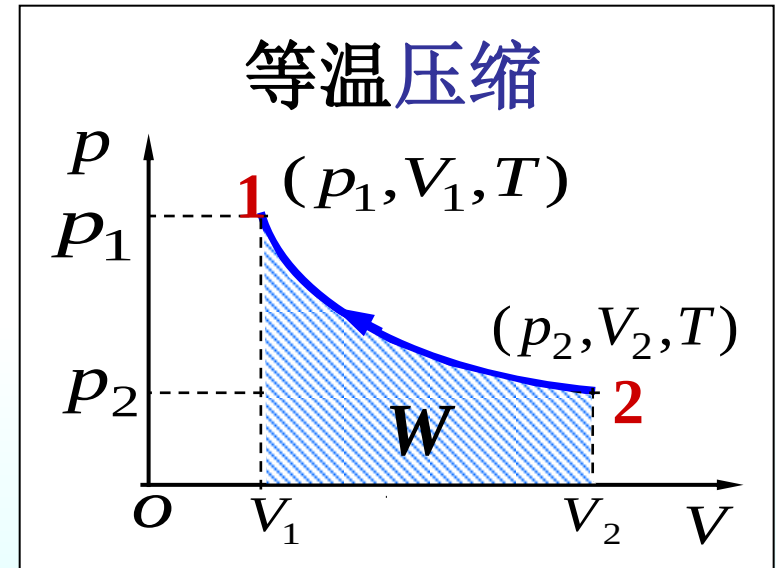
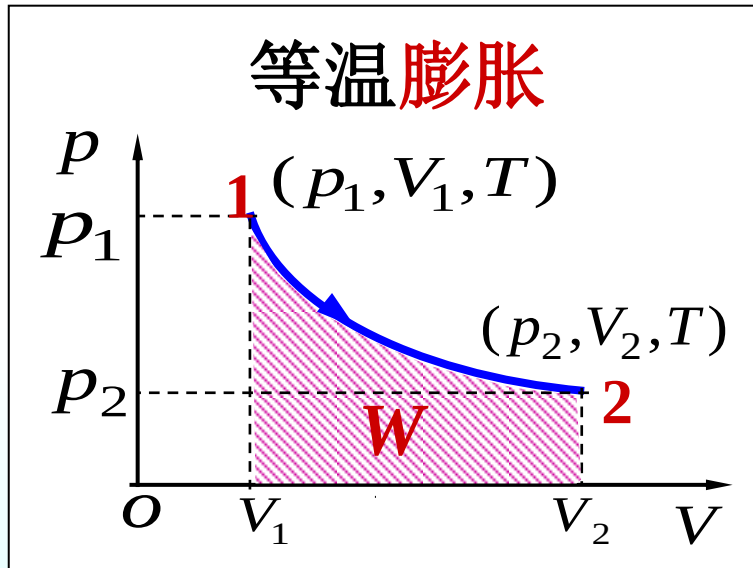


3. Isothermal Processes (等温过程)

(1) Process equation

$$pV = \text{constant}$$

(2) pV diagram



$$(3) \Delta U = 0$$

$$Q_T = W_{12} = \int_{V_1}^{V_2} p dV \quad p = n \frac{RT}{V}$$

$$Q_T = W_{12} = \int_{V_1}^{V_2} n \frac{RT}{V} dV = nRT \ln \frac{V_2}{V_1}$$

$$= nRT \ln \frac{p_1}{p_2}$$

$$(4) C_T = \frac{dQ_T}{dT} = \infty$$

4. **Adiabatic Process** (绝热过程)

(1) Process equation

$$pV^\gamma = C_1 = (const.)$$

$$p^{\gamma-1}T^{-\gamma} = C_2 = (const.)$$

$$V^{\gamma-1}T = C_3 = (const.)$$

$$pV^\gamma = C_1 \quad pV = nRT \Rightarrow p^\gamma V^\gamma = C'T^\gamma$$



There is no time for the gases to exchange heat with their surroundings, so the expansion is adiabatic $Q=0$.

◆ Derive the relation between p and V

$$\because dQ = 0, \quad \therefore dU = -dW = -pdV$$

$$nC_V dT = -pdV \quad (1)$$

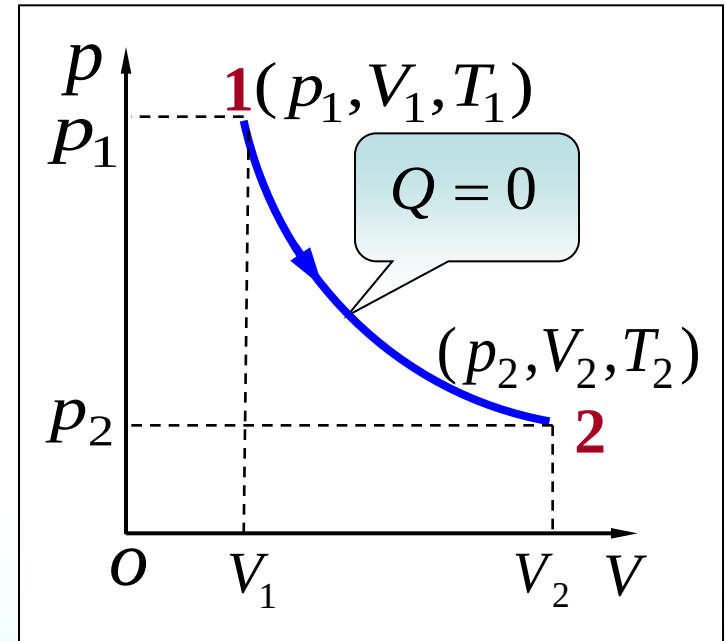
$$pV = nRT$$

$$\Rightarrow pdV + Vdp = nRdT \quad (2)$$

Combine (1) and (2)

$$nC_V \left(\frac{pdV + Vdp}{nR} \right) + pdV = 0$$

$$(C_V + R)pdV + C_V Vdp = 0$$



$$\frac{C_P}{C_V} p dV + V dp = 0 \qquad \gamma = \frac{C_P}{C_V}$$

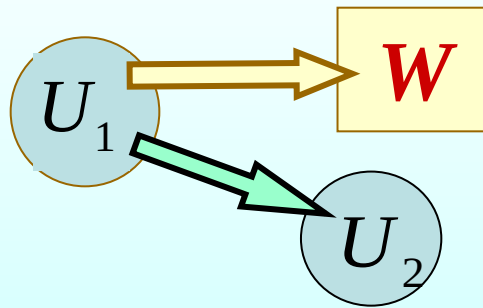
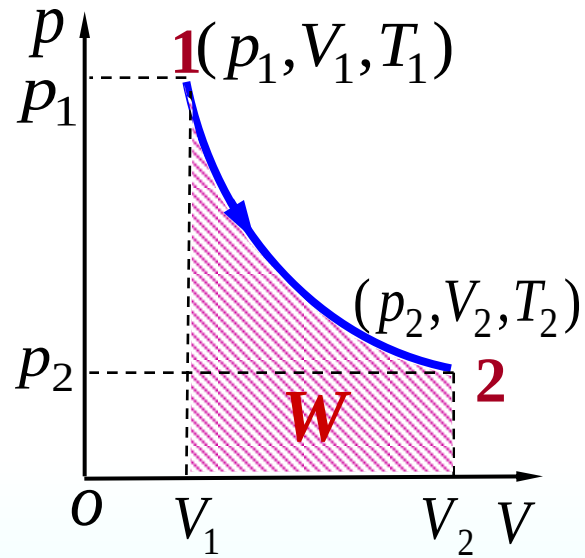
$$\frac{dp}{p} + \gamma \frac{dV}{V} = 0$$

$$\ln p + \gamma \ln V = \text{const}$$

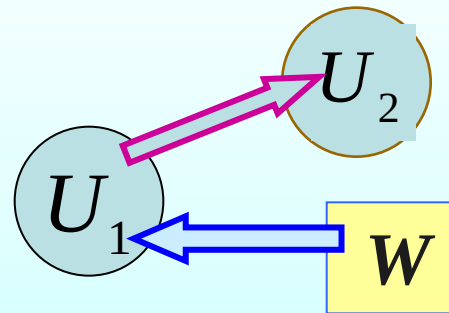
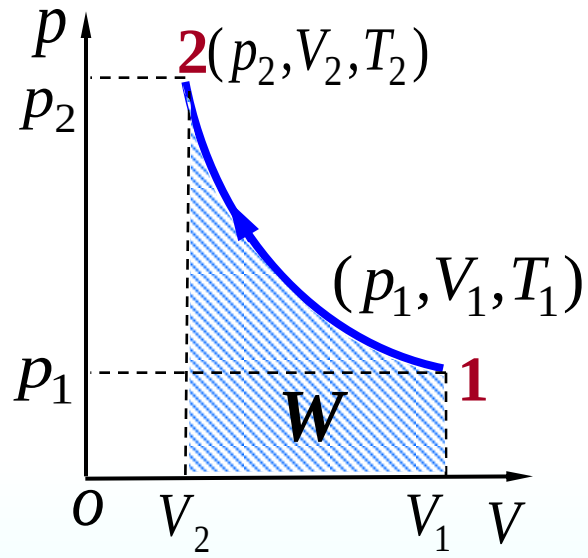
$$pV^\gamma = \text{const}$$

(2) pV diagram

Adiabatic Expansion



Adiabatic Compression



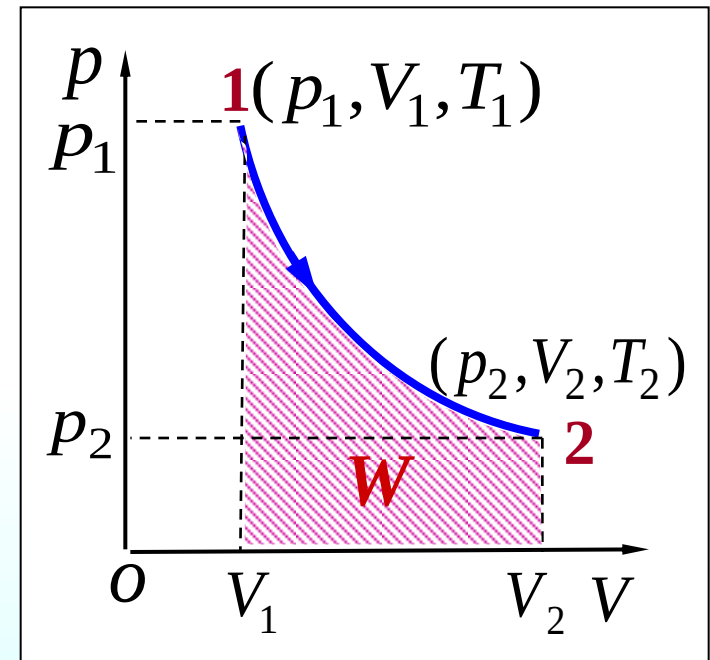
$$(3) W = -\Delta U = -nC_V(T_2 - T_1) = -n \frac{fR}{2} (T_2 - T_1)$$

$$pV = nRT$$

$$W = -\frac{f}{2} (p_2 V_2 - p_1 V_1) = \frac{f}{2} (p_1 V_1 - p_2 V_2)$$

$$\gamma = \frac{f+2}{f} \Rightarrow -\frac{f}{2} = \frac{1}{1-\gamma}$$

$$W = \frac{1}{1-\gamma} (p_2 V_2 - p_1 V_1)$$



Another way

$$W = \int_{v_1}^{v_2} p dV = \int_{v_1}^{v_2} \frac{C_1}{V^r} dV$$

$$= \frac{1}{1-\gamma} \left[\frac{C_1}{V_2^{r-1}} - \frac{C_1}{V_1^{r-1}} \right]$$

$$= \frac{1}{1-\gamma} [p_2 V_2 - p_1 V_1]$$

$$(4) \quad C_a = 0$$

(5) Adiabatic and isotherm

Adiabatic $pV^\gamma = C$

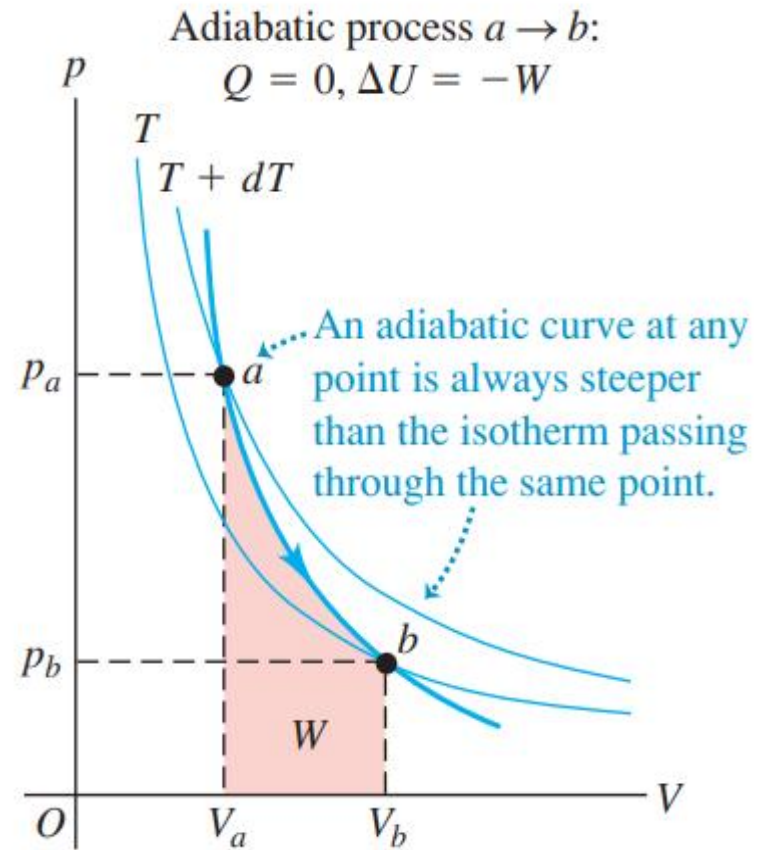
$$\gamma p V^{\gamma-1} dV + V^\gamma dp = 0$$

$$\left(\frac{dp}{dV}\right)_A = -\gamma \frac{p_A}{V_A}$$

isotherm $pV = \text{constant}$

$$\left(\frac{dp}{dV}\right)_T = -\frac{p_A}{V_A}$$

$$pdV + Vdp = 0$$



The PV curve of the adiabat is steeper than the isotherm passing through the same point.

Solving problems

(1) $pV = nRT$ (equation of state)

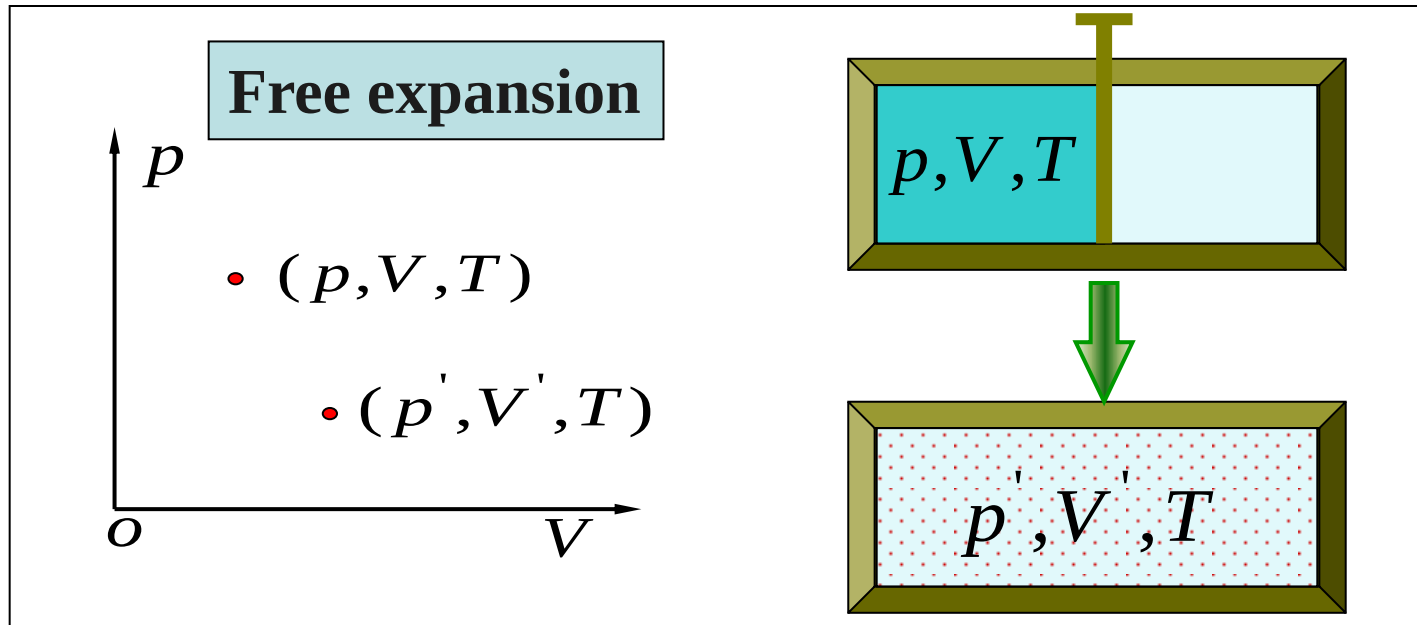
(2)
$$\begin{cases} dU = dQ - pdV \\ \Delta U = Q - \int_{V_1}^{V_2} pdV \end{cases}$$

**The first law of
thermodynamics**

(3) $U = U(T)$ (state quantity)

(4) **Characteristics for different processes**

(adiabatic) free expansion 绝热自由膨胀



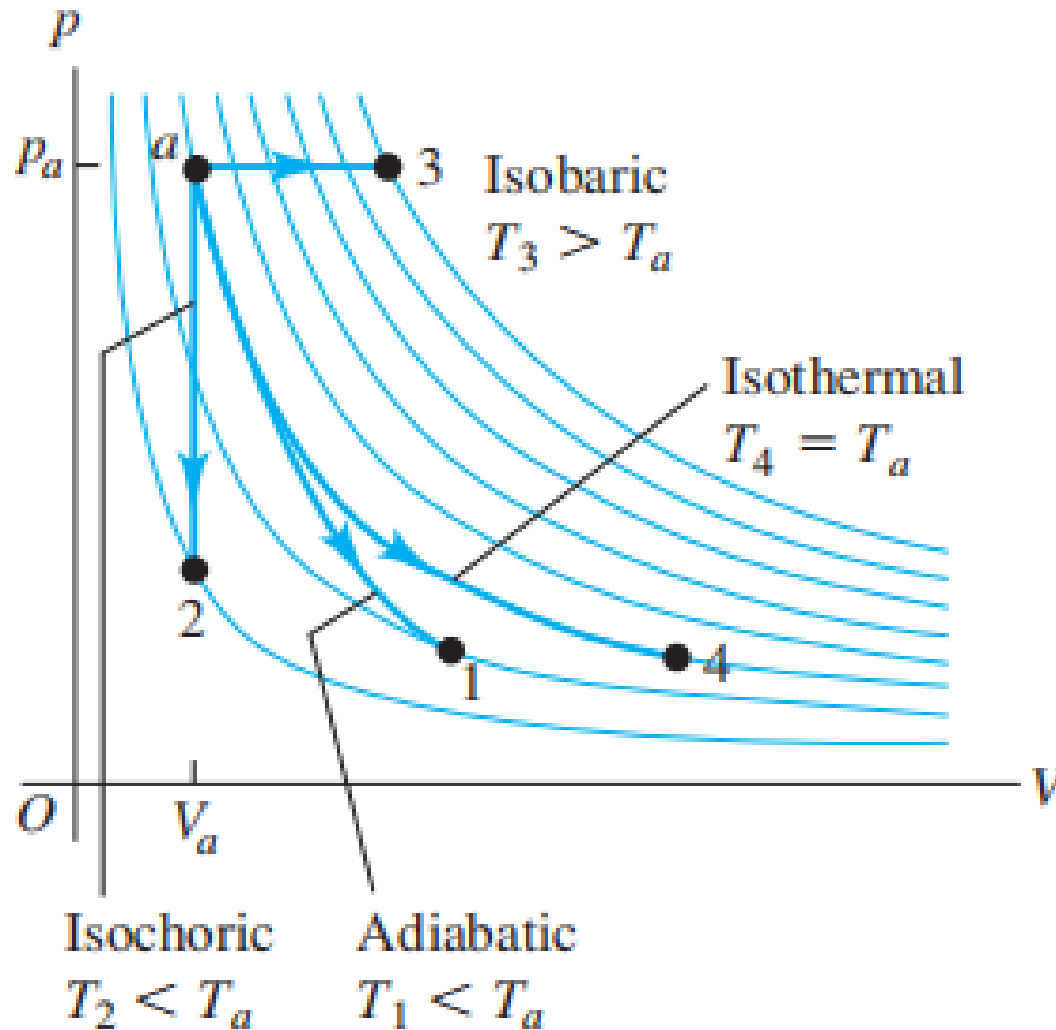
$$Q = 0$$

$W = 0$ (gas does not move any other object)

$$\Delta U = 0$$

$$\Delta T = 0$$

Four different processes for a constant amount of an ideal gas, all starting at state a .



Question : When work W is done on an ideal gas of **diatomic molecules** in **thermal isolation** the increase in the **total rotational energy** of the molecules is:

A. 0

B. $2W/3$

C. $2W/5$

D. $3W/5$

E. W

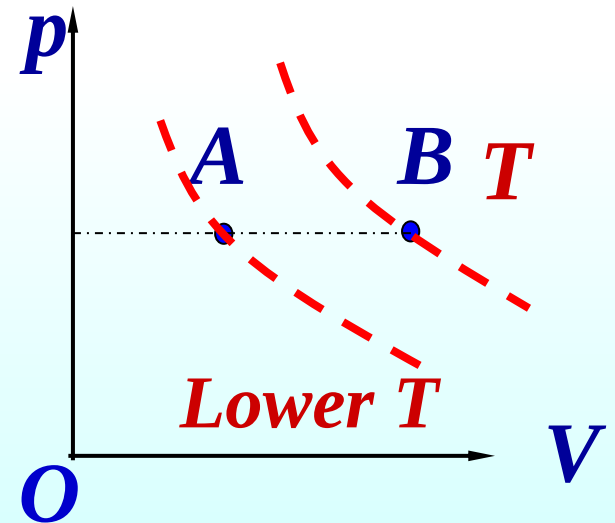
Question: As shown in the figure, a certain amount of the ideal gas goes from equilibrium state A to equilibrium state B at a constant pressure. Then, no matter what process of the gas going through, the gas **surely**

(A) does positive work on environment

(B) increases the internal energy

(C) absorbs heat from environment

(D) gives out heat to environment



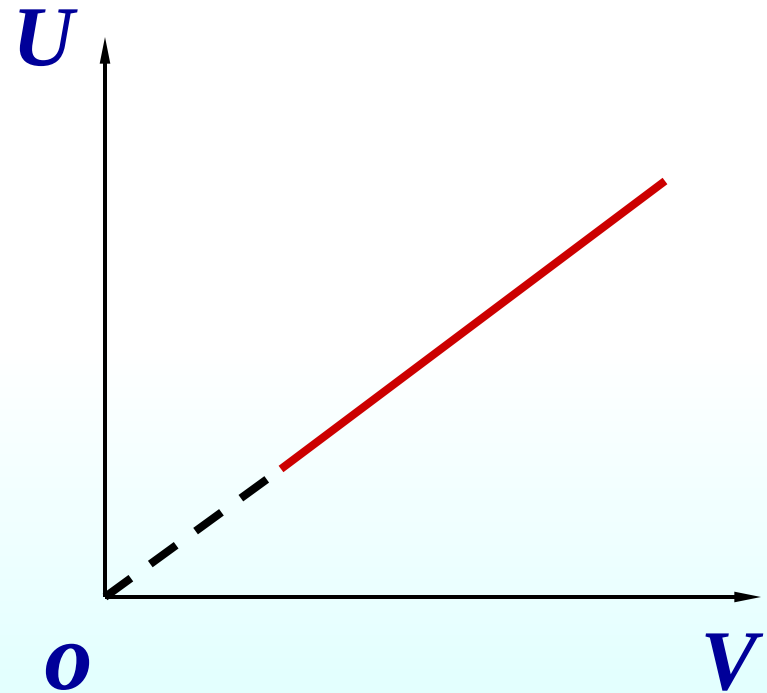
Question: The change in internal energy U of an ideal gas is a function of volume V as shown in the figure, what does the possible process undergo?

Solution:

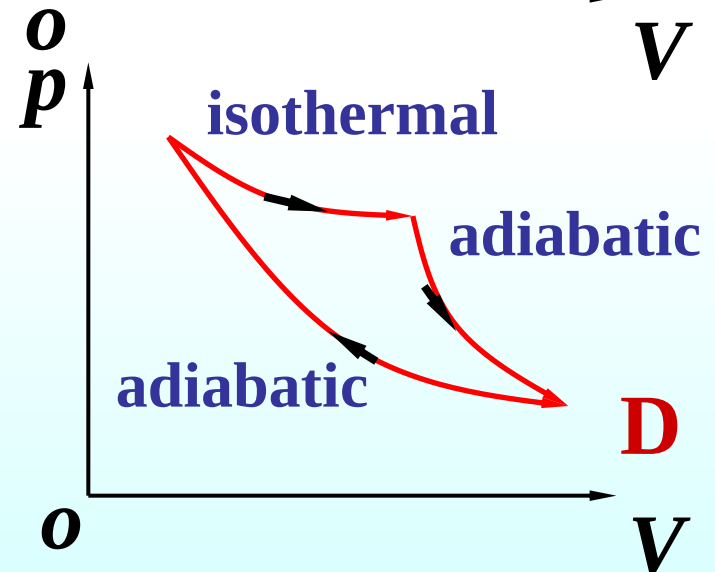
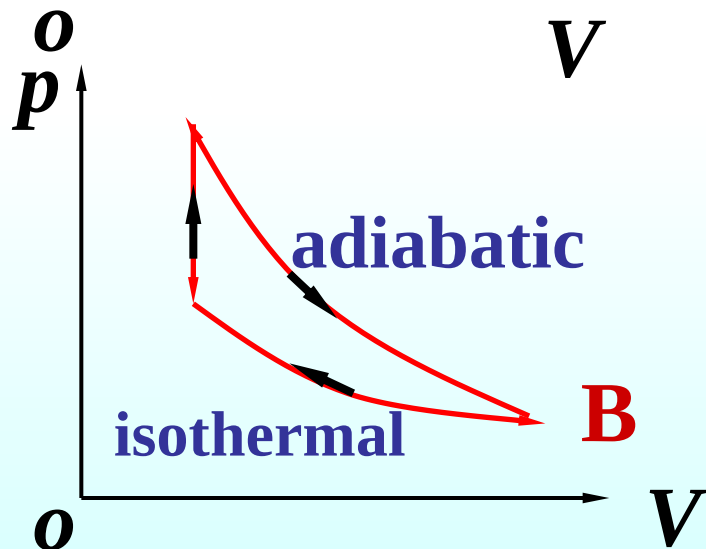
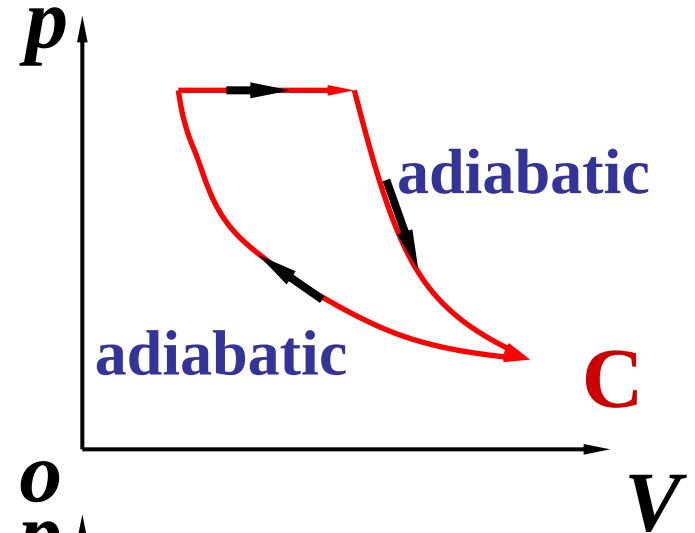
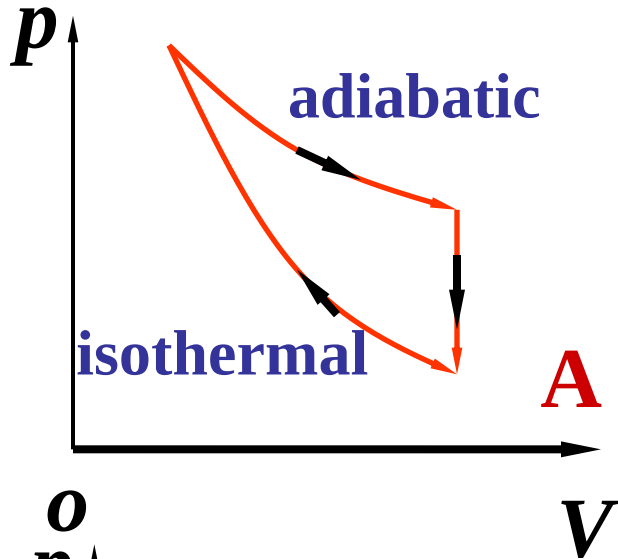
$$U = n f R T / 2 = P V f / 2 = b V$$

$$p = 2b / f$$

Isobaric process



Question : Which of the following four cycles can be achieved in theory?



Example An ideal diatomic gas undergoes process in the figure from state A to state B (P_1 and V_1 are known).

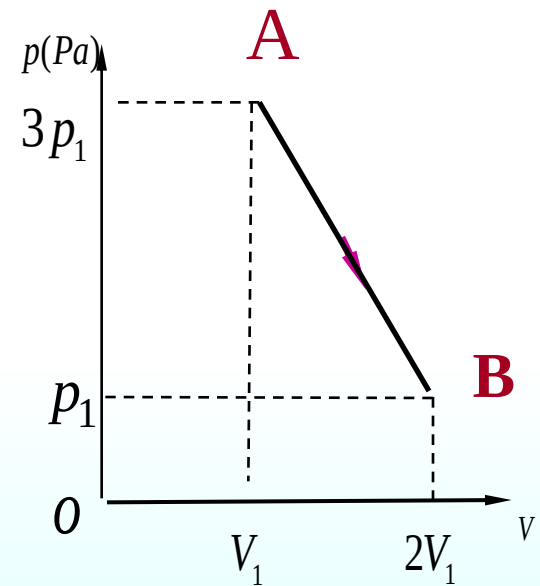
Find the **work done** by the gas and the **change in internal energy**.

Solution: $W = \frac{1}{2}(3p_1 + p_1)(2V_1 - V_1) = 2p_1V_1$

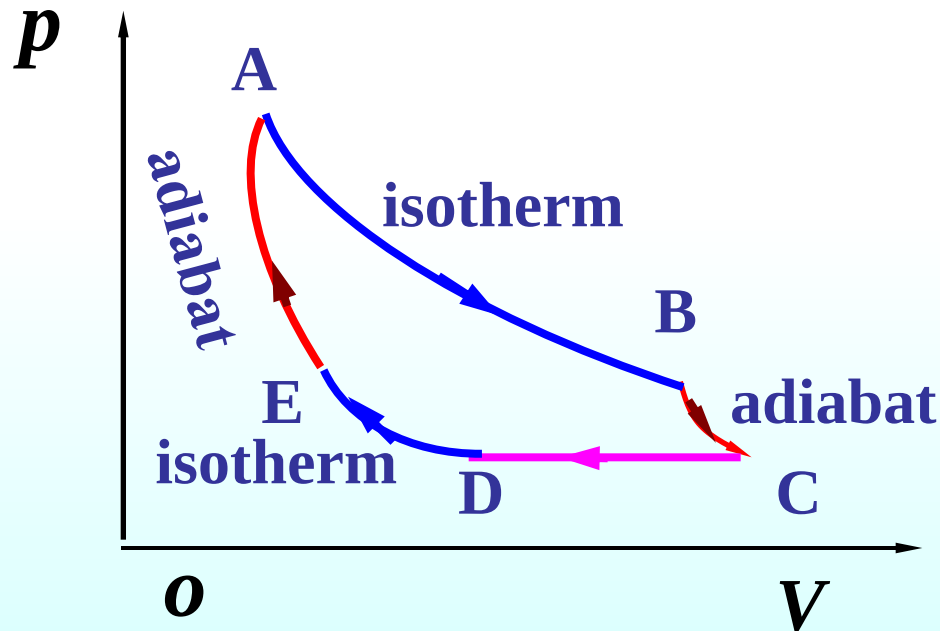
$$T_A = \frac{3p_1V_1}{nR}, T_B = \frac{2p_1V_1}{nR}$$

$$\Delta U = nC_V\Delta T = n\frac{5}{2}R \times (T_B - T_A)$$

$$= n\frac{5}{2}R\left(\frac{2p_1V_1}{nR} - \frac{3p_1V_1}{nR}\right) = -\frac{5}{2}p_1V_1$$

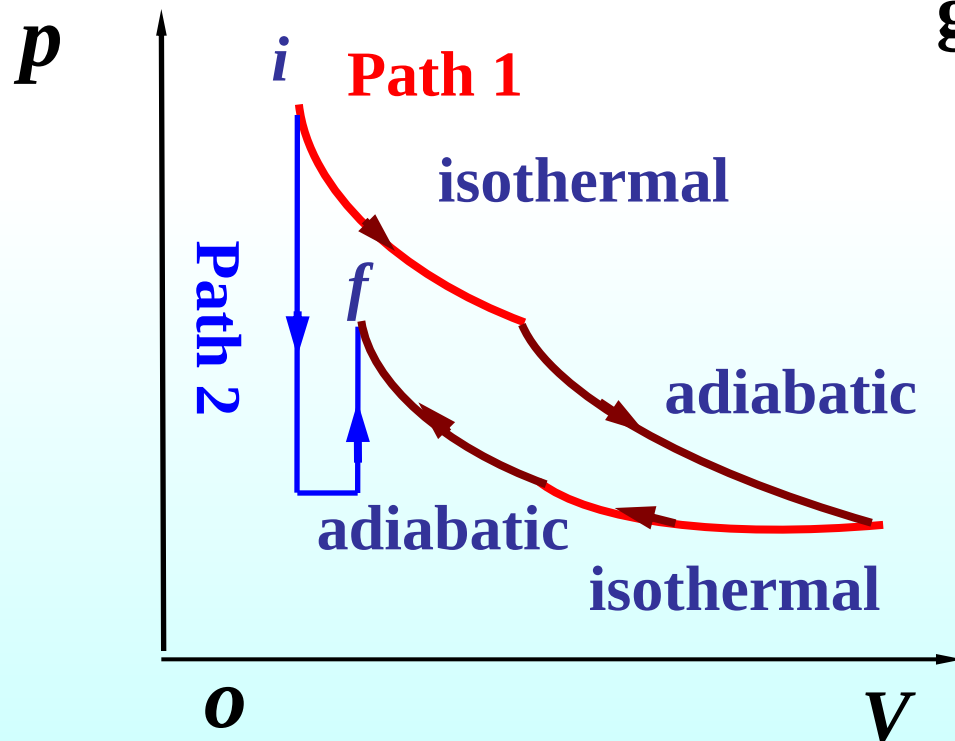


Question: Figure shows a cycle consisting five paths: AB is **isothermal** at 300K, BC is **adiabatic** with work=5.0J, CD is at a constant pressure of 5 atm, DE is **isothermal** and EA is **adiabatic** with a change in internal energy of 8.0J. What is the change in **internal energy** of the gas along path CD?



Question : Figure shows two paths taken from an initial point i to a final point f . Path 1 consists of an isothermal expansion (work is 50J in magnitude), an adiabatic expansion (work is 40J in magnitude), an isothermal compression (work is 30J in magnitude), and then an adiabatic compression (work is 25J in magnitude). What is the change in internal energy of the gas if the gas

goes from point i to point f along path 2?



Example An ideal **diatomic** gas follows two processes in the diagram respectively. (1) it is slowly compressed at a constant pressure of 1.0 atm from 100 cm³ to 20 cm³ and then heat is added to the gas with the volume constant until the temperature reaches its original value (2) it undergoes an isothermal process, state 1 to state 2.

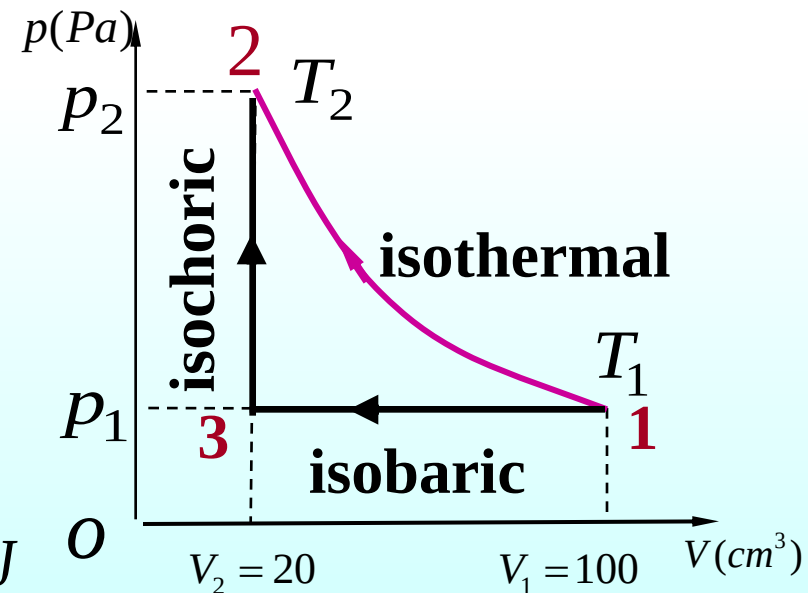
Determine: for two processes (a) the change in internal energy (b) the total heat flow into the gas (c) the total work done by the gas.

Solution: Process 1-2

$$U_1 = U_2, \Delta U = 0$$

$$Q_T = W = nRT \ln \frac{V_2}{V_1} = P_1 V_1 \ln \frac{V_2}{V_1}$$

$$= 1.013 \times 10^5 \times 100 \times 10^{-6} \times \ln \frac{20}{100} = -16.3 \text{ J}$$



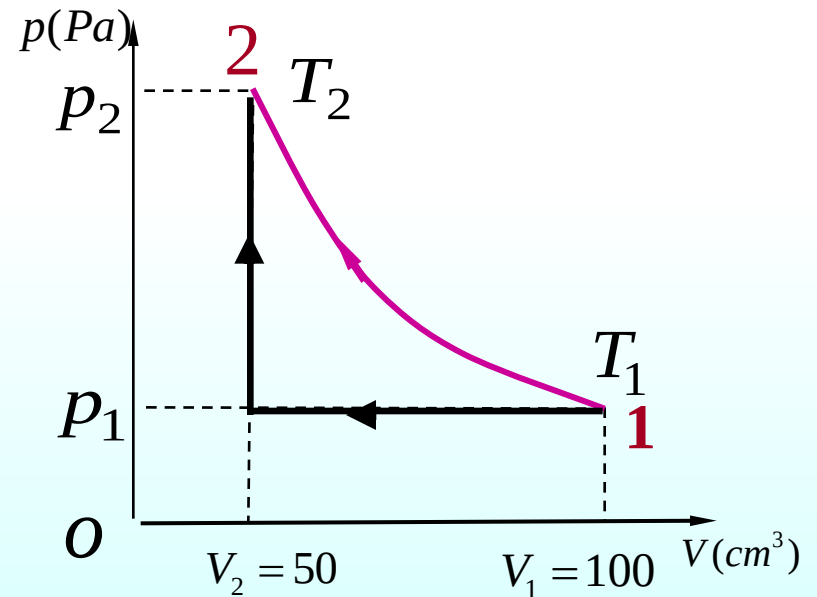
Process 1-3-2

(2) $\Delta U = 0$ (U only depends on the state)

$$\begin{aligned} W &= W_p + W_V = W_p = p_1(V_2 - V_1) \\ &= 1.013 \times 10^5 \times (20 \times 10^{-6} - 100 \times 10^{-6}) = -8.1 J \end{aligned}$$

$$Q = W = -8.1 J$$

Heat and work are process quantities.



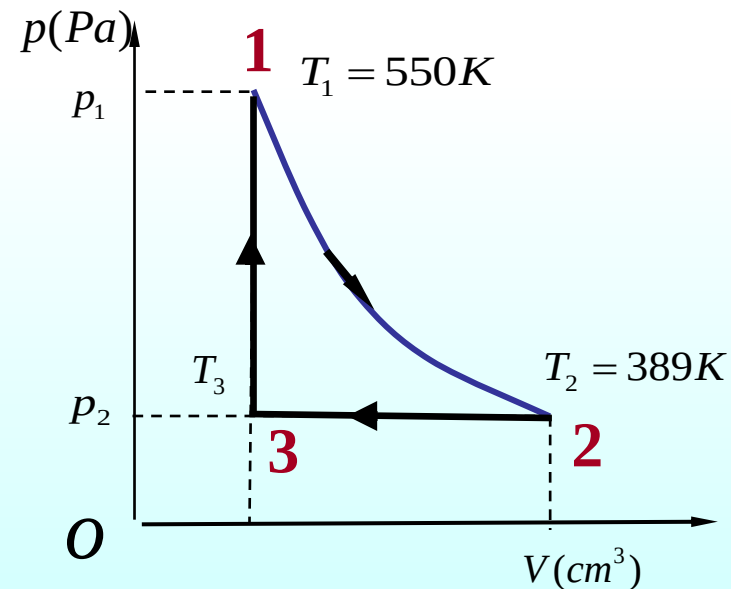
A **1.00-mol** sample of an ideal **monatomic** gas at a pressure of **1.00 atm**, undergoes a three-step process (1) it is expanded **adiabatically** from $T_1=550\text{K}$ to $T_2=389\text{K}$; (2) it is compressed at constant pressure until its temperature reaches T_3 ; (3) it then returns to its original pressure and temperature by a constant volume process. (a) Plot the processes on a PV diagram. (b) Determine T_3 (c) calculate the change in internal energy, the work done by the gas, and the heat added to the gas for each process, and (d) for the complete cycle.

Solution:

State 1: $p_1 V_1 = nR T_1 \Rightarrow V_1 = 0.45\text{m}^3$

1-2 adiabatic $\gamma = \frac{5}{3}$

$$p_1^{\gamma-1} T_1^{-\gamma} = p_2^{\gamma-1} T_2^{-\gamma} \Rightarrow p_2 = 0.42\text{atm}$$



$$p_2 V_2 = nR T_2 \Rightarrow V_2 = 0.76 \text{ m}^3$$

2-3 isobaric

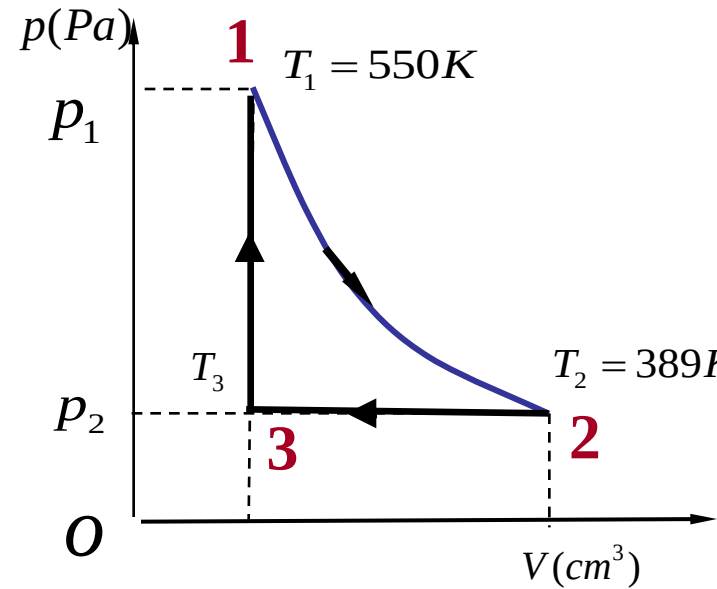
$$p_3 V_3 = nR T_3 \Rightarrow T_3 = p_2 V_1 / nR = 231 \text{ K}$$

(c) Process 1-2 adiabatic

$$Q_{12} = 0$$

$$\Delta U = nC_V(T_2 - T_1) = n \frac{fR}{2} (T_2 - T_1) = \frac{1 \times 3 \times 8.31}{2} (389 - 550) = -2.01 \times 10^3 \text{ J}$$

$$W_{12} = -\Delta U = 2.01 \times 10^3 \text{ J}$$

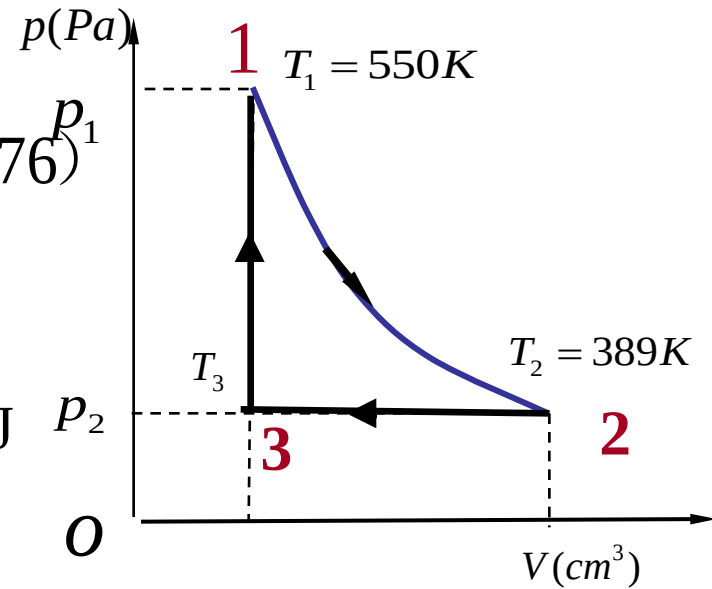


2-3 isobaric

$$W_{23} = p(V_3 - V_2) = 0.42 \times 1.013 \times 10^5 \times (0.45 - 0.76) \\ = -1.31 \times 10^3 \text{ J}$$

$$\Delta U_{23} = nC_V(T_2 - T_1) = n \frac{fR}{2}(T_3 - T_2) = -1.97 \times 10^3 \text{ J}$$

$$Q_{23} = \Delta U + W = -3.28 \times 10^3 \text{ J}$$



3-1 isochoric

$$W_{31} = 0$$

$$Q_{31} = \Delta U_{31} = nC_V(T_1 - T_3) = n \frac{fR}{2}(T_1 - T_3) = 3.98 \times 10^3 \text{ J}$$

(d) For the complete cycle

$$\Delta U = 0$$

$$W_{\text{cycle}} = W_{12} + W_{23} + W_{31} = 0.70 \times 10^3 \text{ J}$$

$$Q_{\text{cycle}} = Q_{12} + Q_{23} + Q_{31} = 0.70 \times 10^3 \text{ J}$$

